AI's Power and Limitation:
a theoretical perspective

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AI's power and limitation

Talk outline:

1. AI's impressive *power*
2. Reflections on *AI's nature*
   -- are there problems *beyond the reach* of AI?
3. AI's new momentum: *AI for Science*
4. New frontier: *AI* ⟺ *Quantum*
5. Anticipating the future

1. Machine Learning as Algorithmic Paradigm

- ML algorithms are being widely used in all domains:
  quality of solutions keep improving empirically,
  although theoretical backing still incomplete.

- Let us reflect on the nature of AI
  to deepen our scientific understanding
  and anticipate future directions.



2. Reflecting on the nature of AI

- Basic machine learning framework:

  Given a task and a model template M_θ (with parameter θ)

  Q1. The Representation Problem

  Does there exist θ such that M_θ solves the task?

  Q2. The Learning Problem

  Can θ be learned in reasonable space/time?

- How is this framework different from CS' algorithmic view?

  In CS the 2 questions are usually treated as 1, using only mathematical analysis without help from data.

- Learning from data adds to the arsenal of algorithm design!

2. Reflecting on the nature of AI (continued)

- AI: Learning from data leads to powerful algorithms:

  playing GO, predicting Protein folding, etc

- Are there limitations to the power of AI?

  Of course, the halting problem remains unsolvable:

  a Turing machine (with augmented data) is a still a Turing machine

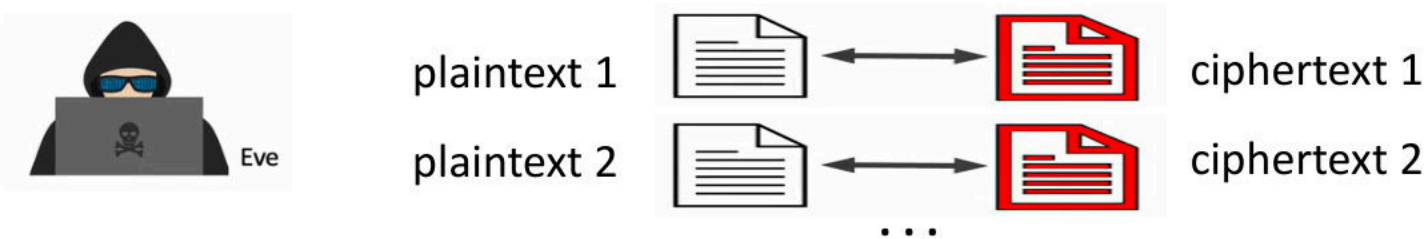
- What practical problems cannot be solved by AI?

  We will give two examples

2. Problems *beyond* the reach of AI

Ex1. Provably secure encryption against AI attackers

- Attack model: Chosen-plaintext attack (CPA)



CPA model captures well a learning-based AI attack

- CPA-security is a vital requirement for public key systems
  (encryption key is public, attacker can use it repeatedly)
- Some crypto schemes, such as El Gamal cipher, is provably secure against chosen plaintext attacks (under DDH hypothesis)
- Cryptography is a rich and deep subject in theoretical CS

Digression: Theoretical Computer Science (TCS)

- TCS started from scratch in the 1960's, and grew organically over the next few decades:

  Fresh ideas popping up - deep theories forming -
  in tandem with advances in computing technology:
  cryptography, internet, AI, quantum computing…

- Interacting with other sciences (physics, biology, economics…), creating new interdisciplinary fields and novel applications

2. Problems *beyond* the reach of AI (continued)

Ex2. Quantum Key Distribution (QKD)

- QKD is a method that allows two parties to create a shared, secret password. It uses the laws of quantum physics to detect eavesdroppers; a hacker's act of measuring the communication will destroy or alter the key, making spying detectable! [Bennett, Brassard 1984] 2025 Turing award

- Implementation of QKD in China:
  - Mozi(墨子号): launched in 2016, the world's first quantum science satellite.
  - QKD infrastructure now supports long-distance key distribution up to 12,800 km, servicing e-government, cross-border finance, power-grid.



3. AI for Science: AI + Astronomy



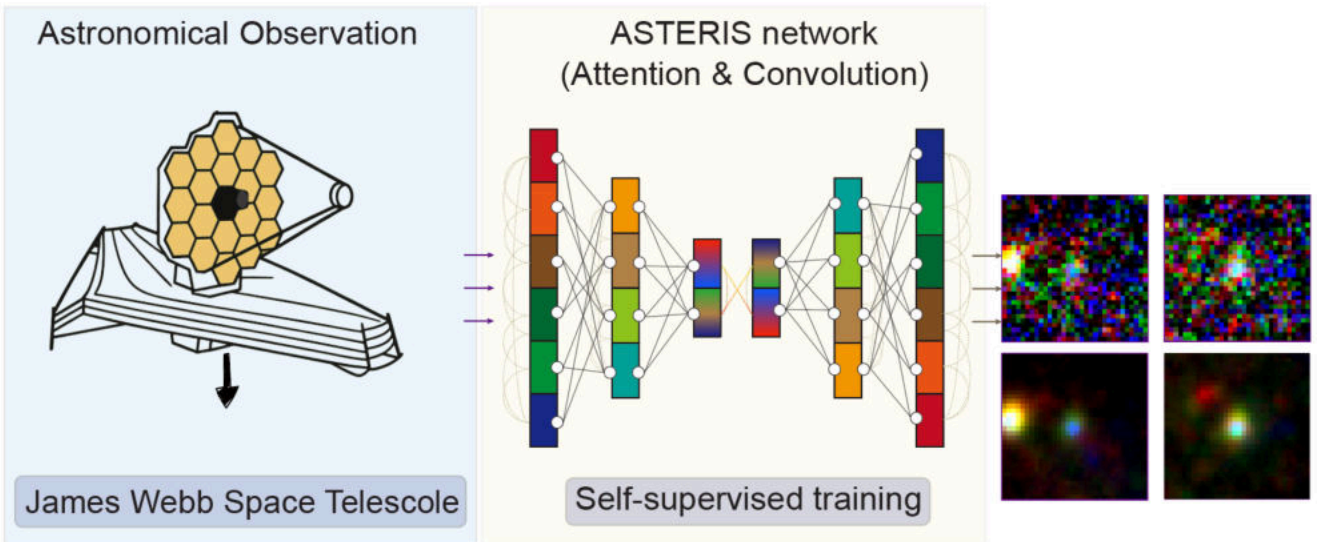
Science, 392 (2026.02)

Y. Guo et al

(Tsinghua Univ)

Example 1 AI+ Astronomy

- Detecting faint galaxies in early universe, seeing stars 100 million years older than state-of-art



3. AI for science: AI + Cosmology

- AI has been used widely in experimental sciences, and recently has begun to make theoretical breakthroughs!

Example 2 Cosmic String Problem (宇宙弦)

One cosmological theory proposes there are "strings" left over from early universe, which still generate gravitational radiation. An open problem is to determine their radiation power spectrum [Garfinlke-Vachaspati, 1987]

(*Relevant for cosmology and outer space exploration*)

- Using Gemini Deep Think, this problem was recently solved with an exact formula. [Brenner, Cohen-Addad, Woodruff 2026]

  * Theoretical computer scientists are playing a strong role in the recent *AI for science* drive.

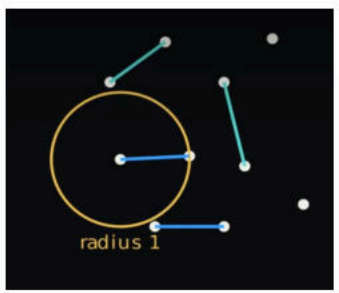


3. AI for science: AI + Math

- AI has been used widely in experimental sciences, and recently has begun to make theoretical breakthroughs!

Example 3 Erdős famous *Unit Distance Problem*

What's the max number of pairs of points $f(n)$ that can be exactly distance 1 apart, in a set of n points in 2D? [problem posed in 1946]



- $f(3)=3$, $f(4)=5$ …

  Erdős conjectured that $f(n)$ is essentially linear
- It is a prominent mathematical open problem over the last 80 years
- Conjecture disproved by OpenAI reasoning model in May 2026, proof technique uses deep algebraic number theory!

3. AI for science: AI + Math

- A newest stunning case:

- Example 4 A Clay Millenium Problem *Naviers-Stokes Equation*

- Does smooth, well-behaved solution always exist? Or show opposite
- Recent work by OpenAI model (and separately by Buckmaster/Alpoge) provides evidence that finite blowup solution happens

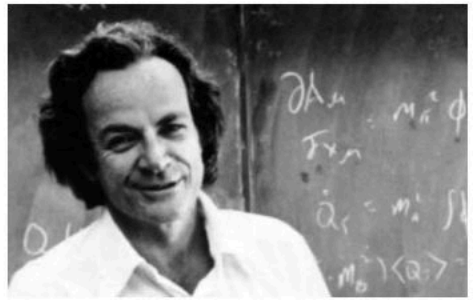
(AI proofs still undergoing human verification.)

$$\partial \mathbf{u}/\partial t + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nu \nabla \mathbf{u} + \mathbf{f}(\mathbf{x}, \mathbf{t})$$

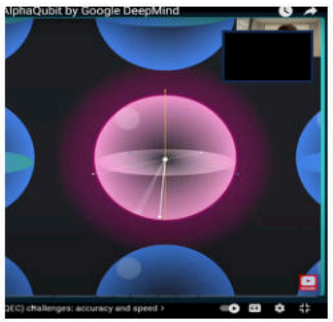
$$\nabla \cdot \mathbf{u} = 0 \qquad (x \in R\,\hat{}\,3,\ t \geq 0)$$

4. New frontier: *AI* ⟺ *Quantum*

1981: Richard Feynman proposed quantum computers!



- Quantum bits (qbit) uses superposition, different from the classical 0-1 bit
- Galvanized research by physicists/computer scientists:
  - Peter Shor (1993): Quantum computers can solve some hard problems (like crypto) beyond classical computers
  - Peter Shor (1996): If quantum error falls below some threshold value (< 1%), then we can realize scalable quantum computation!
  - Quantum error correction has been a major challenge for building quantum computers until recently …
    AI coming to the help!



4. New frontier: *AI* ⟺ *Quantum*

Building Quantum Computers

- qbit is susceptible to interference from environment changing its value – error correction is critical



Quantum chip Willow (by Google) [Nature 2024.12]

- using 105 qbit/chip to reduce error rate to ~ 0.143%
- First time meeting "below < 1%" quantum error threshold challenge
- A landmark step in materializing scalable quantum computation

4. New frontier: *AI* ⟺ *Quantum*

Quantum chip Willow (Google)

How is its error correction achieved?

Core technique:

"AlphaQubit" is a decoder designed for quantum error correction.

- It is a neural network using a huge number of quantum simulation samples, obtained through machine learning training.
- Big Open Problem: Improvements needed in #qbits, accuracy, and speed.

Follow-up work progressing well...

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Learning high-accuracy error decoding for quantum processors

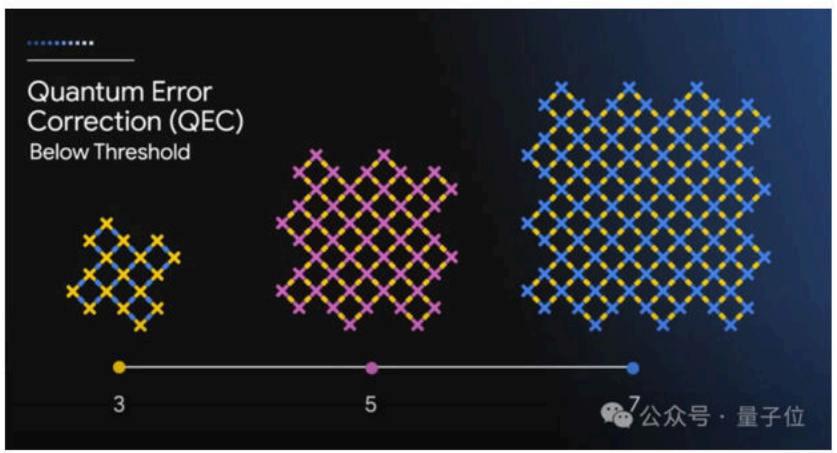
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Abstract

Building a large-scale quantum computer requires effective strategies to correct errors that



4. New frontier: *AI* ⟺ *Quantum*

AI ⟺ Quantum is 2-way empowerment:

"⟹" AI enabling Quantum error correction breakthrough

"⟸" Quantum computer may well expand AI's power

Quantum AI:

- Use quantum machine to learn from data (even quantum data)
- An emerging frontier beyond present-day AI, potentially even more powerful

5. Anticipating the Future

- AI has demonstrated vast power across many domains
- AI algorithms still work within boundaries set by physical laws and theoretical analysis, which provide foundation for designing secure AI systems.
- Much exciting exploration lies ahead, including novel AI for Science directions such as:
  - Quantum AI a frontier beyond present-day AI, potentially more powerful
  - Reliable (even provable) large AI systems
  - AI Safety: involving cryptography, game theory, social sciences
  - AI for AI: deeper understanding of the nature of intelligence
- Can AI solve the P versus NP Problem (another Clay Millenium Problem)?

Thanks!